

RIVER/STREAM RESTORATION

Hydro-ecological Engineering

Understanding the flood pulse concept

- Spawning
- Habitat creation
- Food supply
- Improved water quality

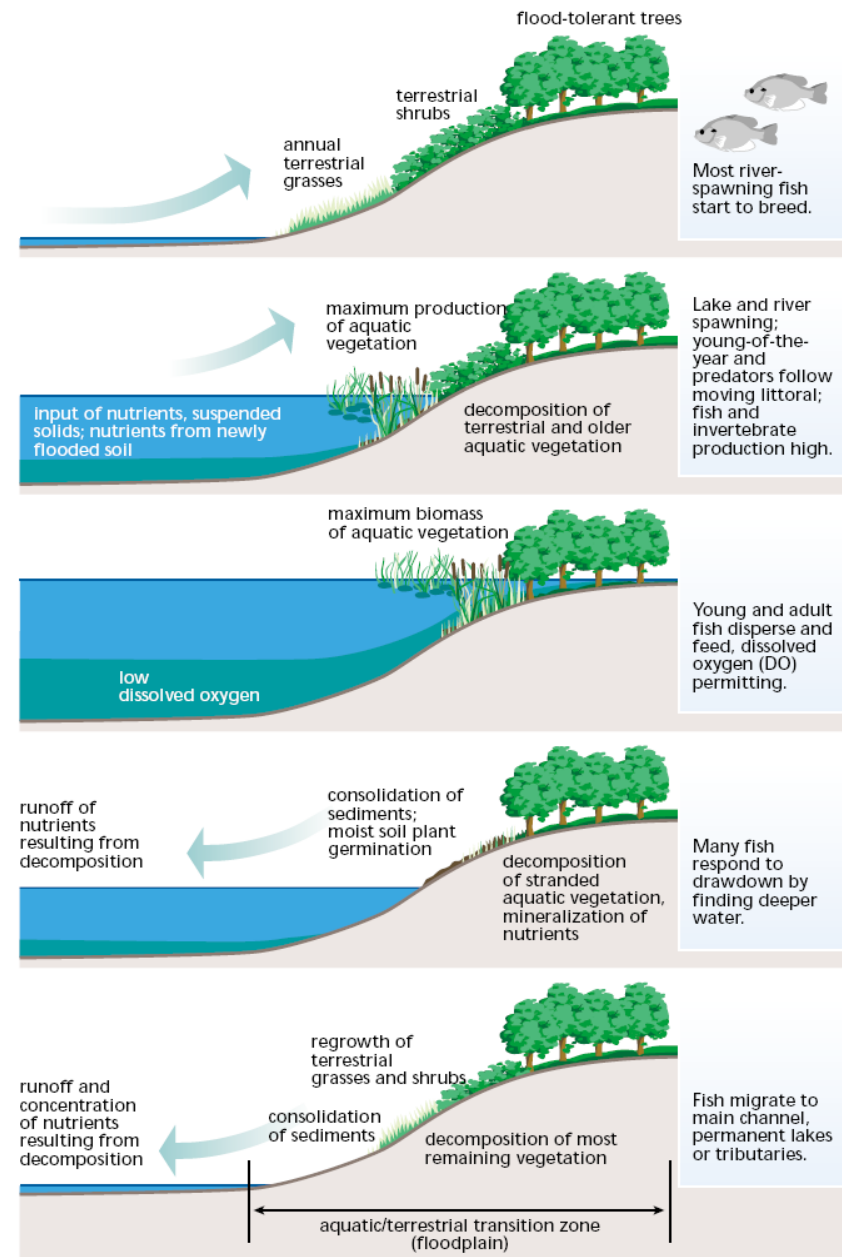


Figure 1.26: Schematic of the flood-pulse concept. A vertically exaggerated section of a floodplain in five snapshots of an annual hydrological cycle. The left column describes the movement of nutrients. The right column describes typical life history traits of fish.

Source: Bayley, *Bioscience*, vol. 45, p.154, March 1995. ©1995 American Institute of Biological Science.

- *Restoring physical habitat cannot restore biological integrity of a system if there are water quality constraints on the ecosystem.*
- *Restoration activities may interact in a variety of complex ways with water quality, affecting both the delivery and impact of water quality stressors.*

Understanding the nitrogen cycle

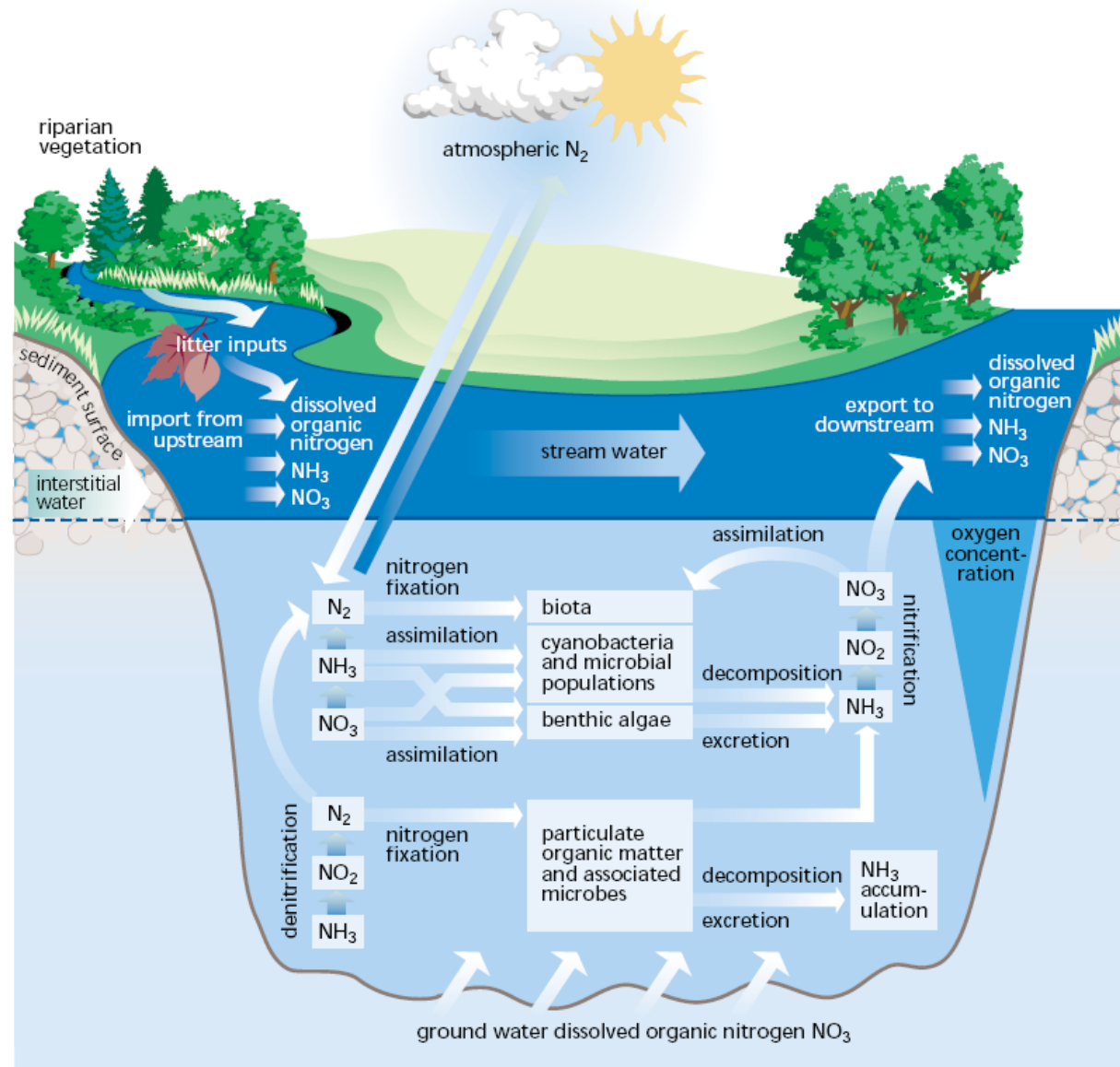


Figure 2.21: Dynamics and transformations of nitrogen in a stream ecosystem. Nutrient cycling from one form to another occurs with changes in nutrient inputs, as well as temperature and oxygen available.

Table 2.6: Sources and concentrations of pollutants from common point and nonpoint sources.

Source	Total Nitrogen (mg/L)	Total Phosphorus (mg/L)
Urban runoff ^a	3–10	0.2–1.7
Livestock operations ^a	6–800 ^b	4–5
Atmosphere (wet deposition) ^a	0.9	0.015 ^c
90% forest ^d	0.06–0.19	0.006–0.012
50% forest ^d	0.18–0.34	0.013–0.015
90% agriculture ^d	0.77–5.04	0.085–0.104
Untreated wastewater ^a	35	10
Treated wastewater ^{a,e}	30	10

^a Novotny and Olem (1994).

^b As organic nitrogen.

^c Sorbed to airborne particulate.

^d Omernik (1987).

^e With secondary treatment.

Table 2.9: Potential water quality impacts of selected stream restoration and watershed management practices.

Restoration Activities	Fine Sediment Loads	Water Temperature	Salinity	pH	Dissolved Oxygen	Nutrients	Toxics
Reduction of land-disturbing activities	Decrease	Decrease	Decrease	Increase/decrease	Increase	Decrease	Decrease
Limit impervious surface area in the watershed	Decrease	Decrease	Negligible effect	Increase	Increase	Decrease	Decrease
Restore riparian vegetation	Decrease	Decrease	Decrease	Decrease	Increase	Decrease	Decrease
Restore wetlands	Decrease	Increase/decrease	Increase/decrease	Increase/decrease	Decrease	Increase	Increase
Stabilize channel and restore under-cut banks	Decrease	Decrease	Decrease	Decrease	Increase	Decrease	Negligible effect
Create drop structures	Increase	Negligible effect	Negligible effect	Increase/decrease	Increase	Negligible effect	Decrease
Reestablish riffle substrate	Negligible effect	Negligible effect	Negligible effect	Increase/decrease	Increase	Negligible effect	Negligible effect

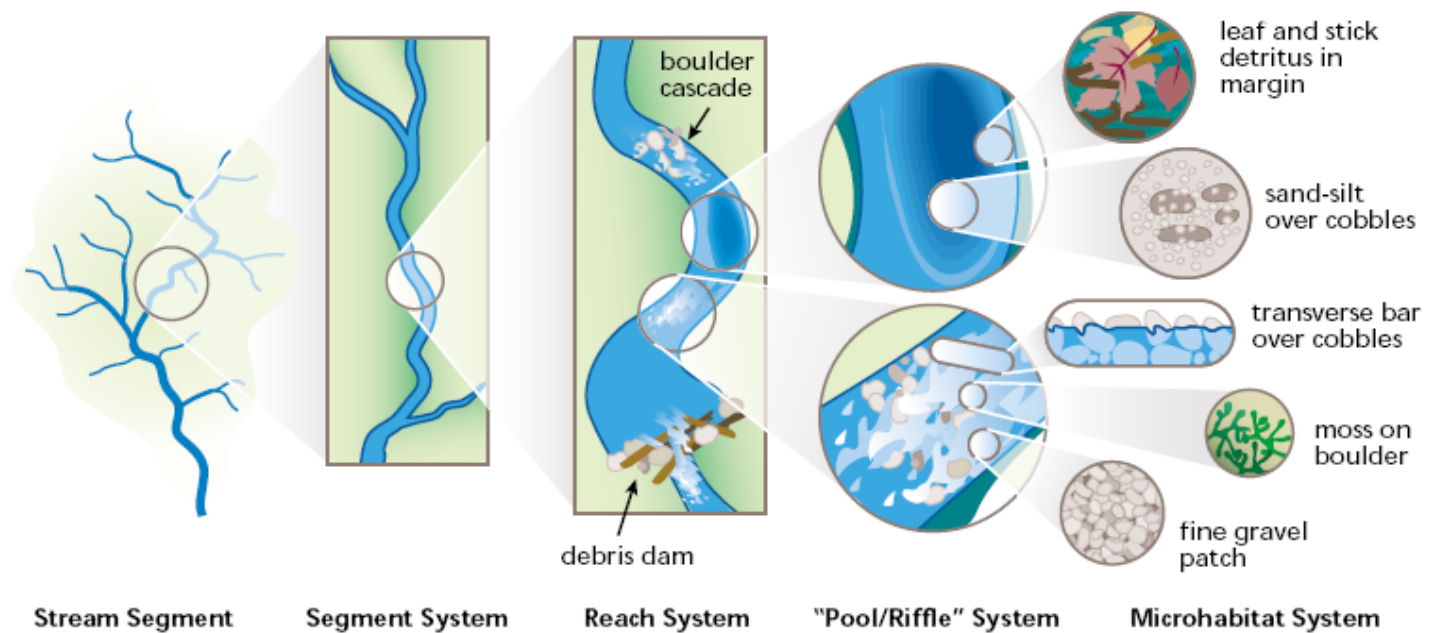


Figure 2.32: Hierarchical organization of a stream system and its habitat subsystems.
Approximate linear spatial scale, appropriate to second- or third-order mountain stream.

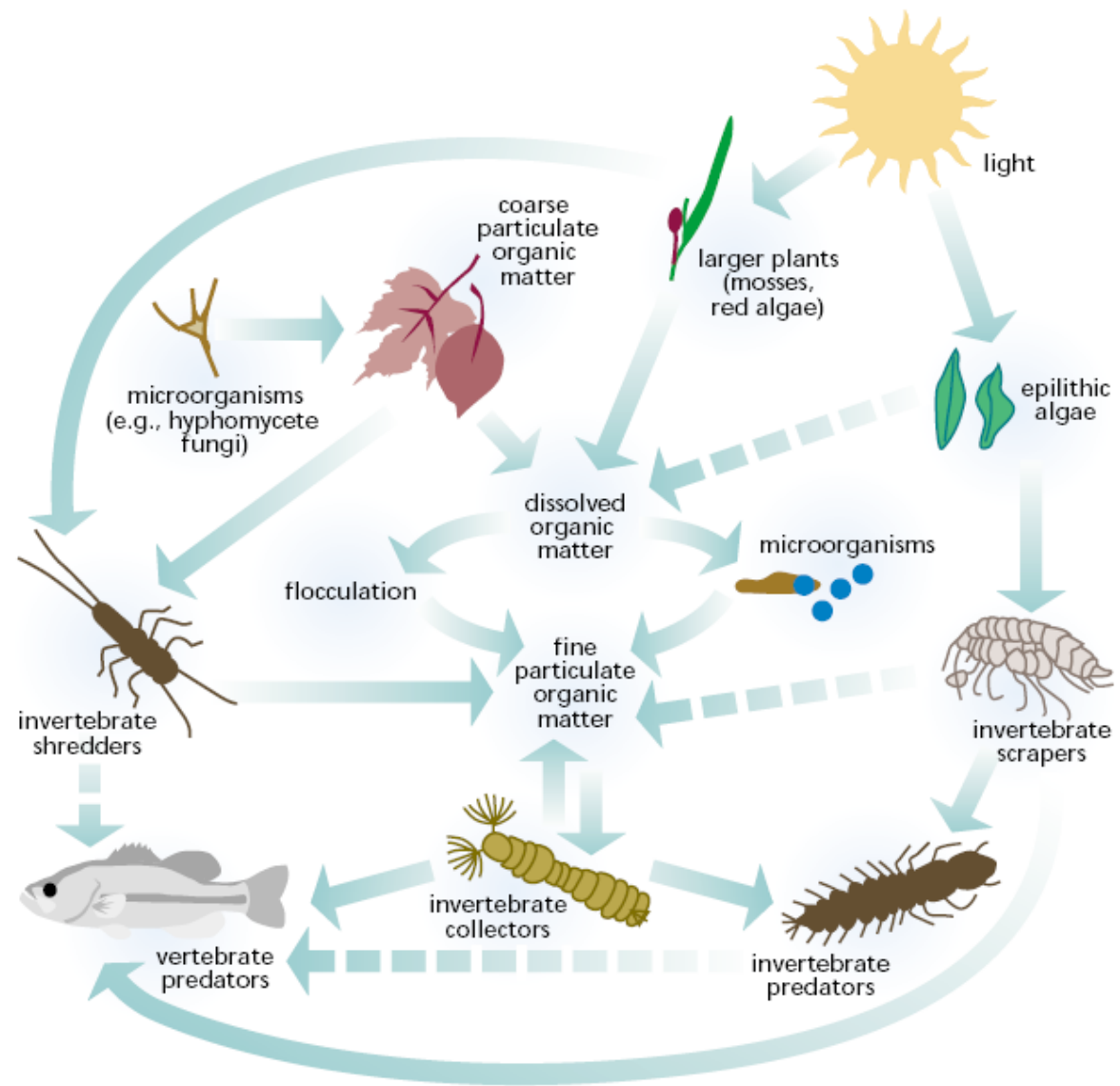
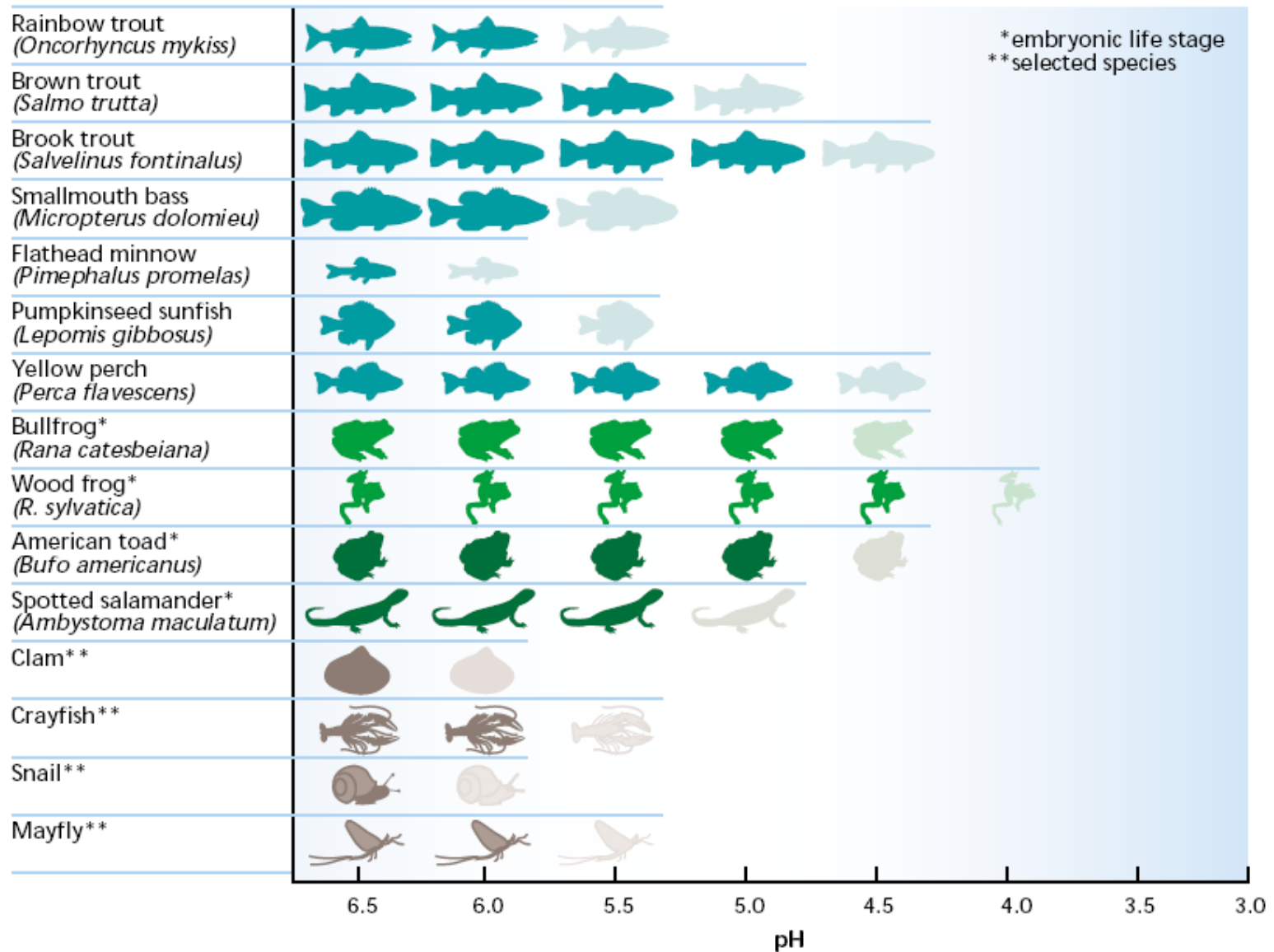


Figure 2.33: Stream biota. Food relationships typically found in streams.

Table 2.12: Ranges of densities commonly observed for selected groups of stream biota.

Biotic Component	Density (Individuals/Square Mile)
Algae	$10^9 - 10^{10}$
Bacteria	$10^{12} - 10^{13}$
Protists	$10^8 - 10^9$
Microinvertebrates	$10^3 - 10^5$
Macroinvertebrates	$10^4 - 10^5$
Vertebrates	$10^0 - 10^2$

Figure 2.34: Effects of acid rain on some aquatic species. As acidity increases (and pH decreases) in lakes and streams, some species are lost.



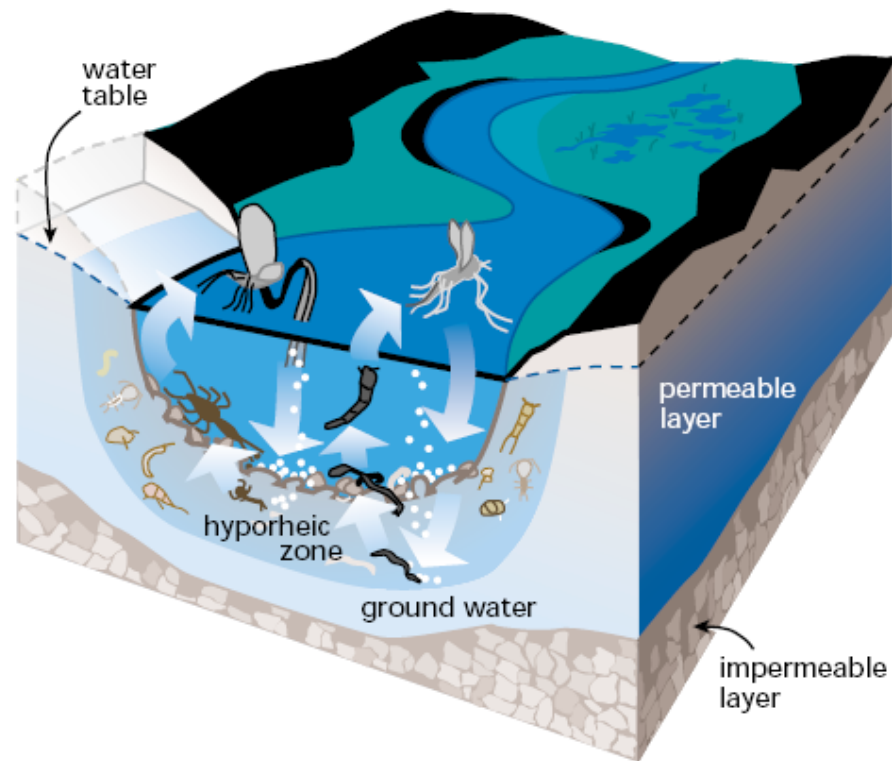
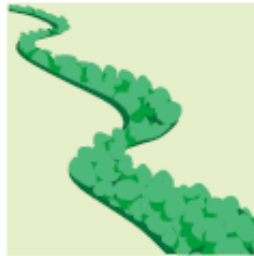


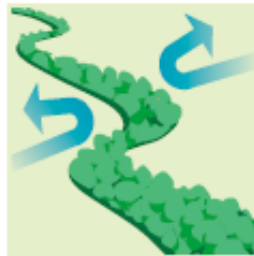
Figure 2.35: Hyporheic zone. Summary of the different means of migration undergone by members of the stream benthic community.

Habitat—the spatial structure of the environment which allows species to live, reproduce, feed, and move.



Habitat

Barrier—the stoppage of materials, energy, and organisms.



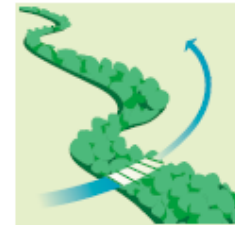
Barrier

Conduit—the ability of the system to transport materials, energy, and organisms.



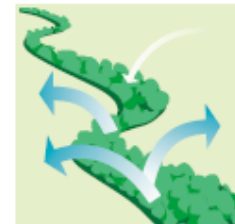
Conduit

Filter—the selective penetration of materials, energy, and organisms.



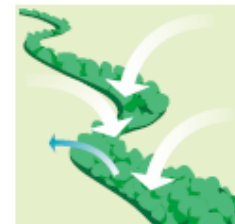
Filter

Source—a setting where the output of materials, energy, and organisms exceeds input.



Source

Sink—a setting where the input of water, energy, organisms and materials exceeds output.



Sink

Figure 2.37: Critical ecosystem functions. Six functions can be summarized as a set of basic, common themes recurring in a variety of settings.

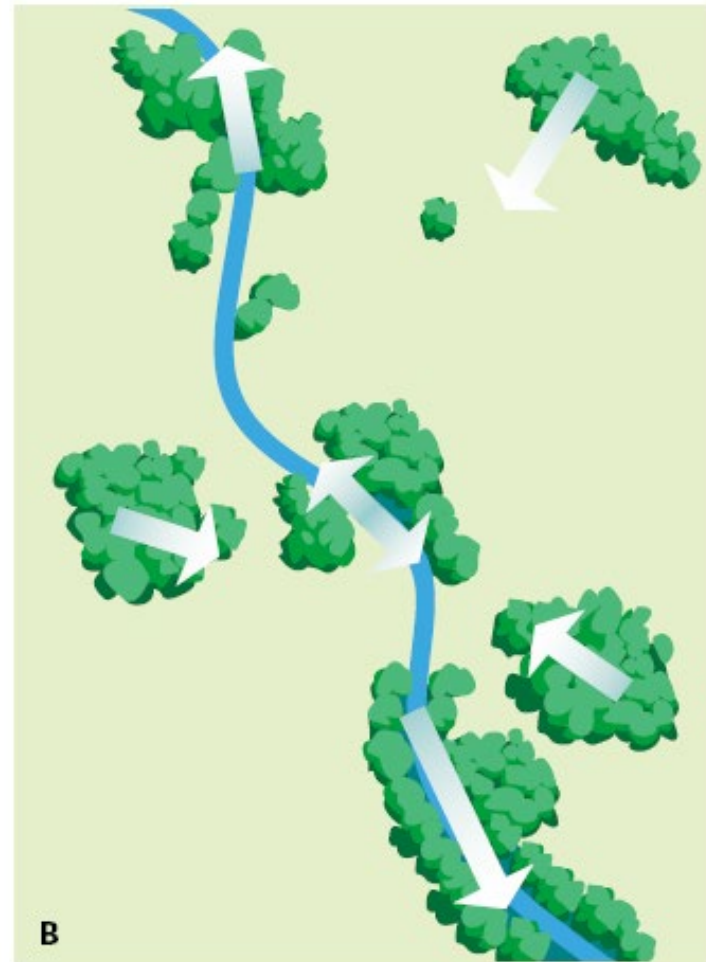
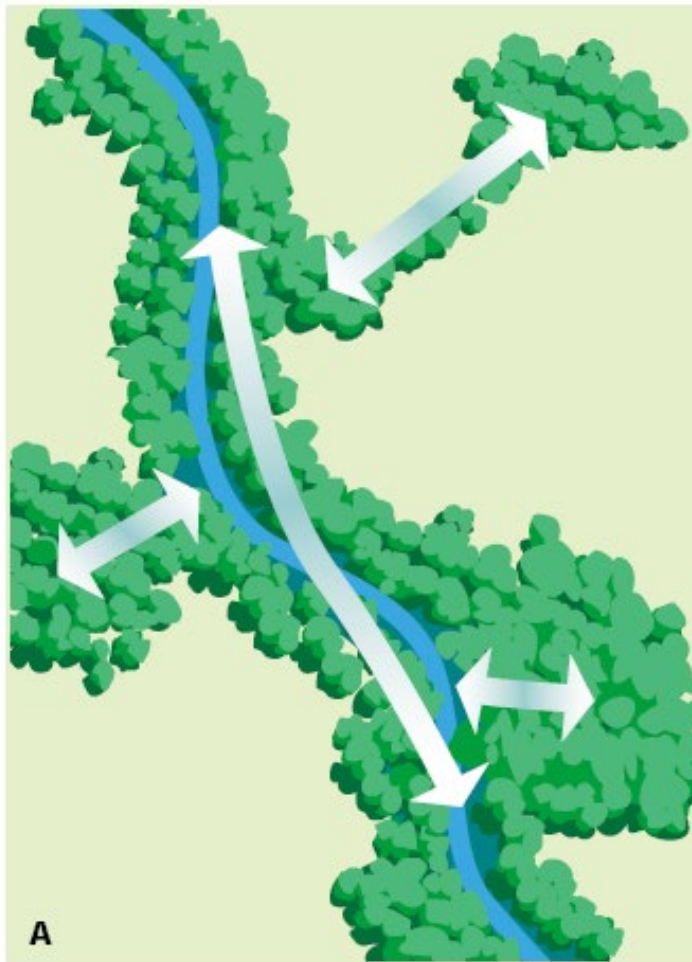


Figure 2.38: Landscapes with (A) high and (B) low degrees of connectivity. A connected landscape structure generally has higher levels of functions than a fragmented landscape.

Vegetation buffers

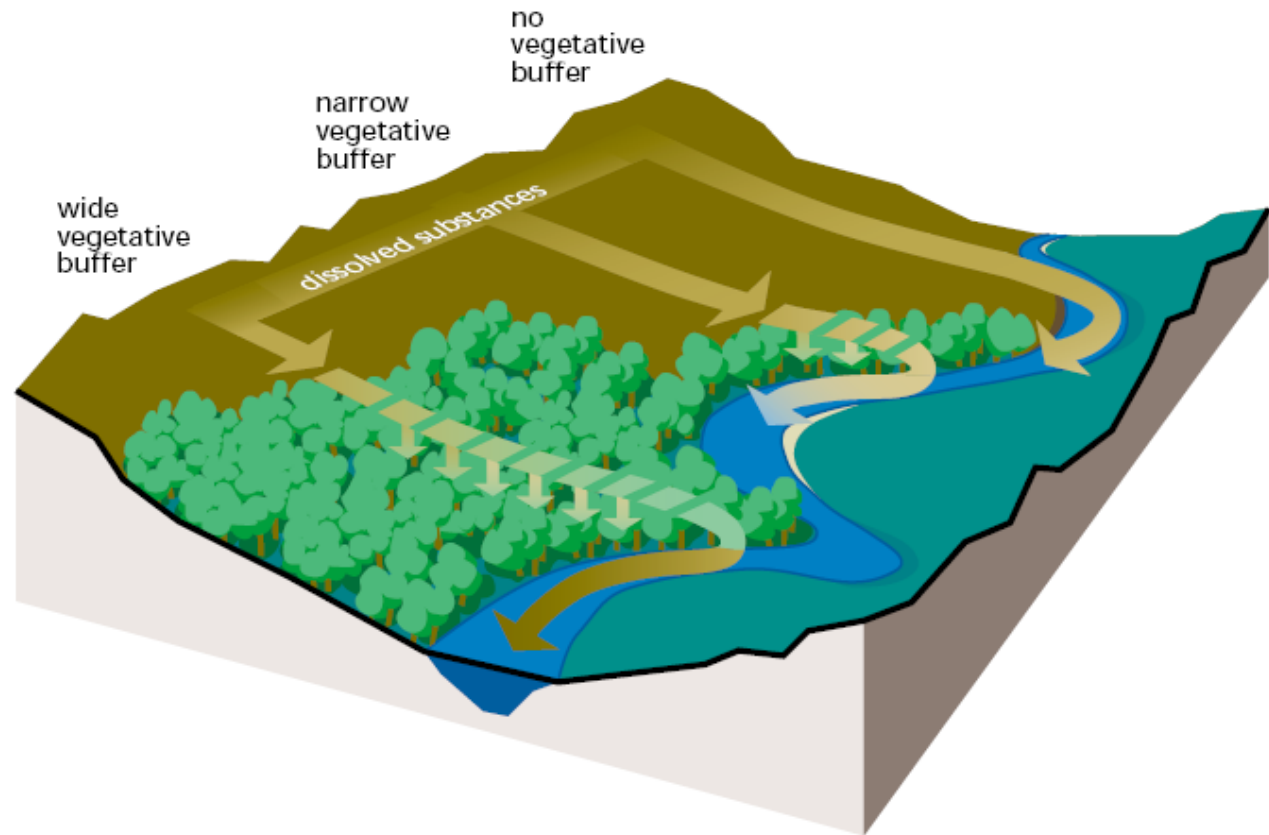


Figure 2.40: *The width of the vegetation buffer influences filter and barrier functions. Dissolved substances, such as nitrogen, phosphorus, and other nutrients, entering a vegetated stream corridor are restricted from entering the channel by friction, root absorption, clay, and soil organic matter.*

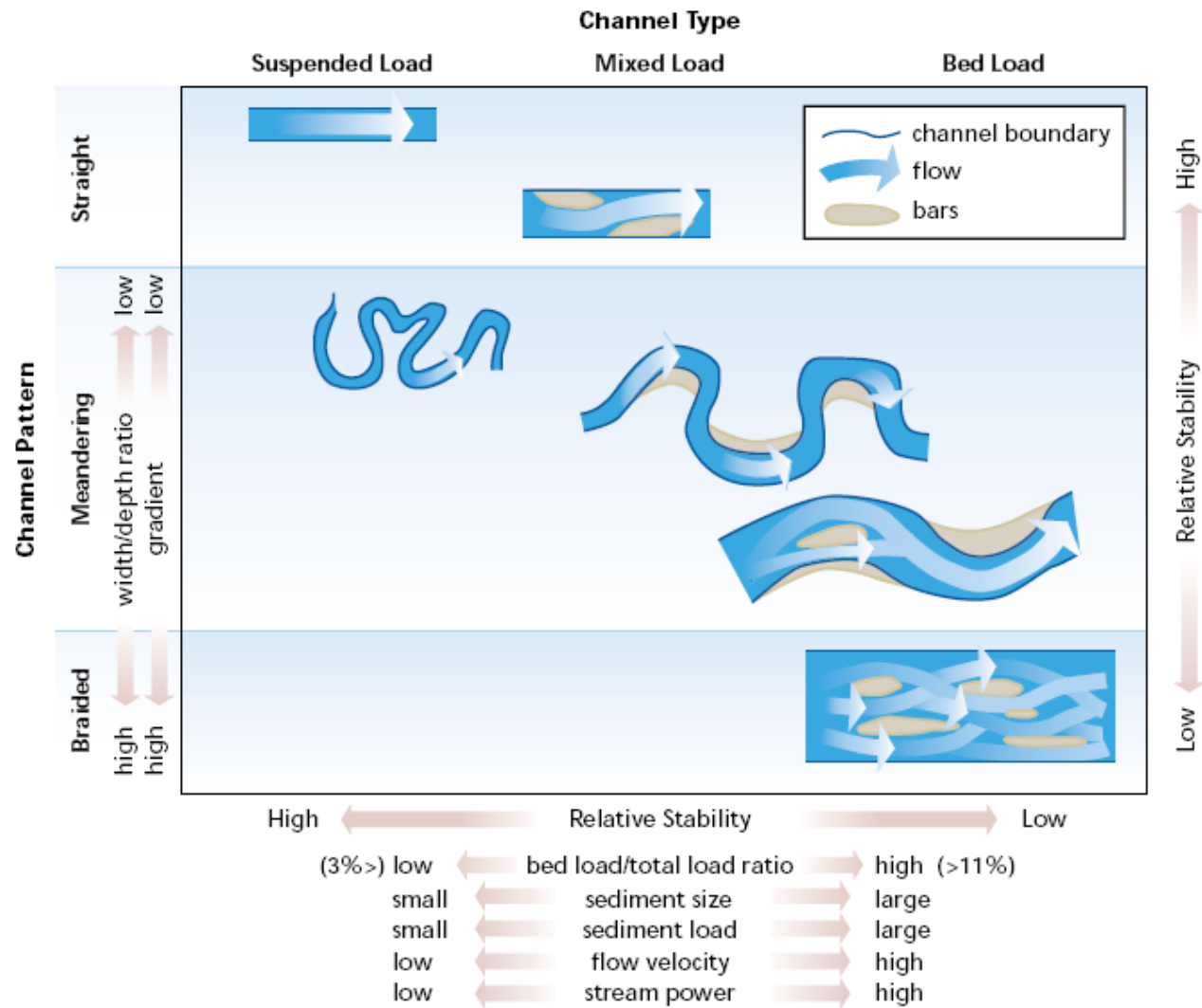
Adapted from Ecology of Greenways: Design and Function of Linear Conservation Areas.

Edited by Smith and Hellmund. © University of Minnesota Press 1993.

Design Discharge for ecological purposes

Design Discharge and Ecological Function

Although a channel-forming or dominant discharge is important for design, it is often not sufficient for channel restoration initiatives. An assessment of a wider range of discharges might be necessary to ensure that the functional objectives of the project are met. For example, a restoration initiative targeting low-flow habitat conditions must consider the physical conditions in the channel during low flows.



Understanding Stream Corridor Dynamics

- Fluxes of energy, water and materials that occur throughout the stream corridor system create a dynamic three-dimensional (length, width, depth) mosaic of different habitats and physical features

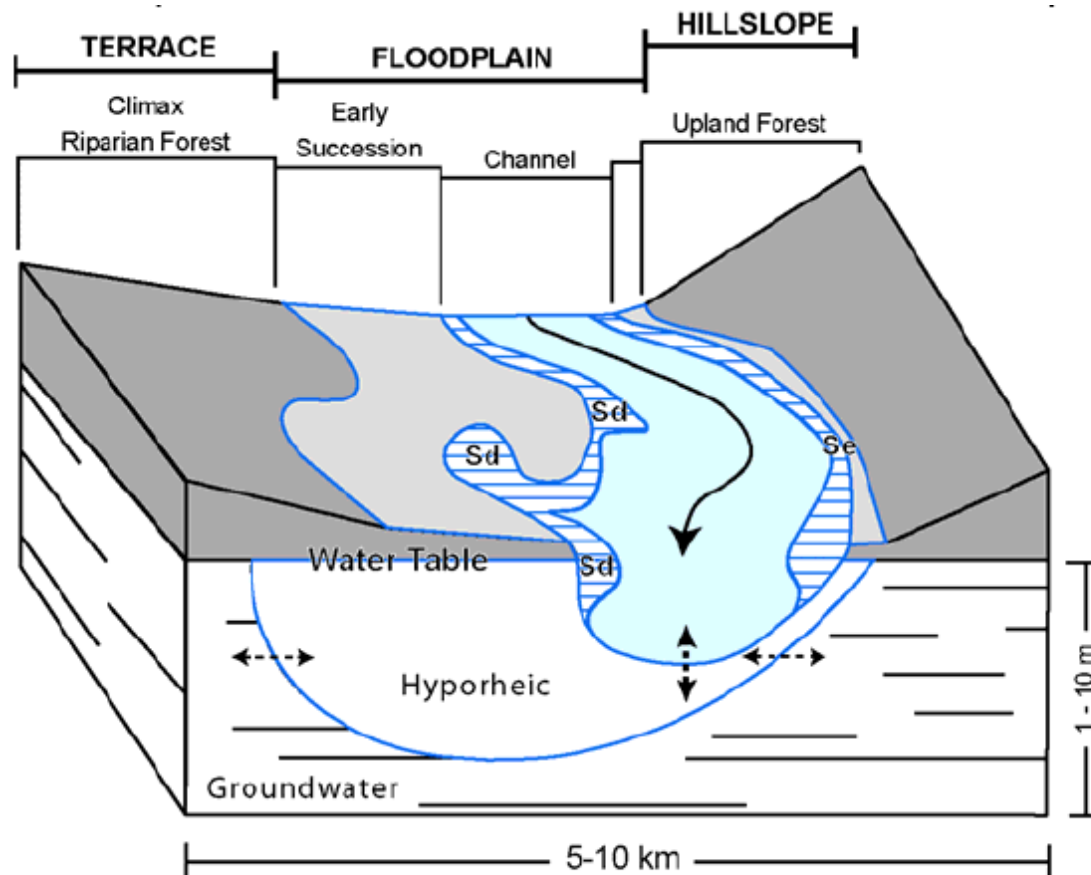


Figure 1-1. Cross-sectional view of a generalized stream corridor segment, showing the three spatial dimensions in which stream corridor habitats are formed through time.

Stream & Floodplain Features

lateral migration

cutbank

meander belt width

point bar

abandoned meander (cutoff)

meander scrolls

Stream & Floodplain Features

lateral migration

cutbank

point bar

meander belt width

abandoned meander (cutoff)

meander scrolls

This aerial photograph illustrates various geomorphic features of a river meander system. The main channel of the river is shown in a sharp curve. A dashed yellow line, labeled 'lateral migration', runs parallel to the outer bank of the curve, indicating the path of the river's movement over time. A 'cutbank' is visible on the outer bank, where the river has eroded the land. A 'point bar' is located on the inner bank, where sediment has accumulated. The 'meander belt width' is the area between the dashed lines. An 'abandoned meander (cutoff)' is shown as a separate, isolated loop of the river. 'Meander scrolls' are the smaller, tighter loops of the river channel. A yellow arrow points to the 'point bar'.

An aerial photograph of a river meandering through a landscape of agricultural fields. The river forms a large loop. Two dashed yellow lines parallel to the main river channel define the 'meander belt width'. A yellow arrow points to a 'cutbank' on the outer curve of the meander. Another yellow arrow points to a 'point bar' on the inner curve. A yellow arrow points to a 'meander scroll' in the abandoned meander (cutoff) loop below the main meander. The title 'Stream & Floodplain Features' is in a blue box at the top left.

Stream & Floodplain Features

lateral migration

cutbank

meander belt width

point bar

abandoned meander (cutoff)

meander scrolls

Stream & Floodplain Features

lateral migration

cutbank

point bar

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abandoned meander (cutoff)

meander scrolls

This aerial photograph illustrates various geomorphic features of a river meander system. The main river channel is shown in a meandering pattern. A dashed line indicates the lateral migration of the channel. A cutbank is visible on the outer bank of a meander, and a point bar is located on the inner bank. The meander belt width is the area between the dashed lines. An abandoned meander (cutoff) is shown as a separate loop. Meander scrolls are the small, crescent-shaped features along the outer bank of a meander.

Stream & Floodplain Features

lateral migration

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Stream & Floodplain Features

lateral migration

cutbank

point bar

meander belt width

abandoned meander (cutoff)

meander scrolls

This aerial photograph illustrates various geomorphic features of a river meander system. The main channel is shown with a prominent meander loop. A dashed line indicates the lateral migration of the channel. A cutbank is visible on the outer bank of the meander, and a point bar is located on the inner bank. The meander belt width is the area between the dashed lines. An abandoned meander (cutoff) is shown as a scroll-like feature. Meander scrolls are the small, crescent-shaped features at the tips of the meander loops.

White River Just Downstream of Soldier Creek Confluence (NE)

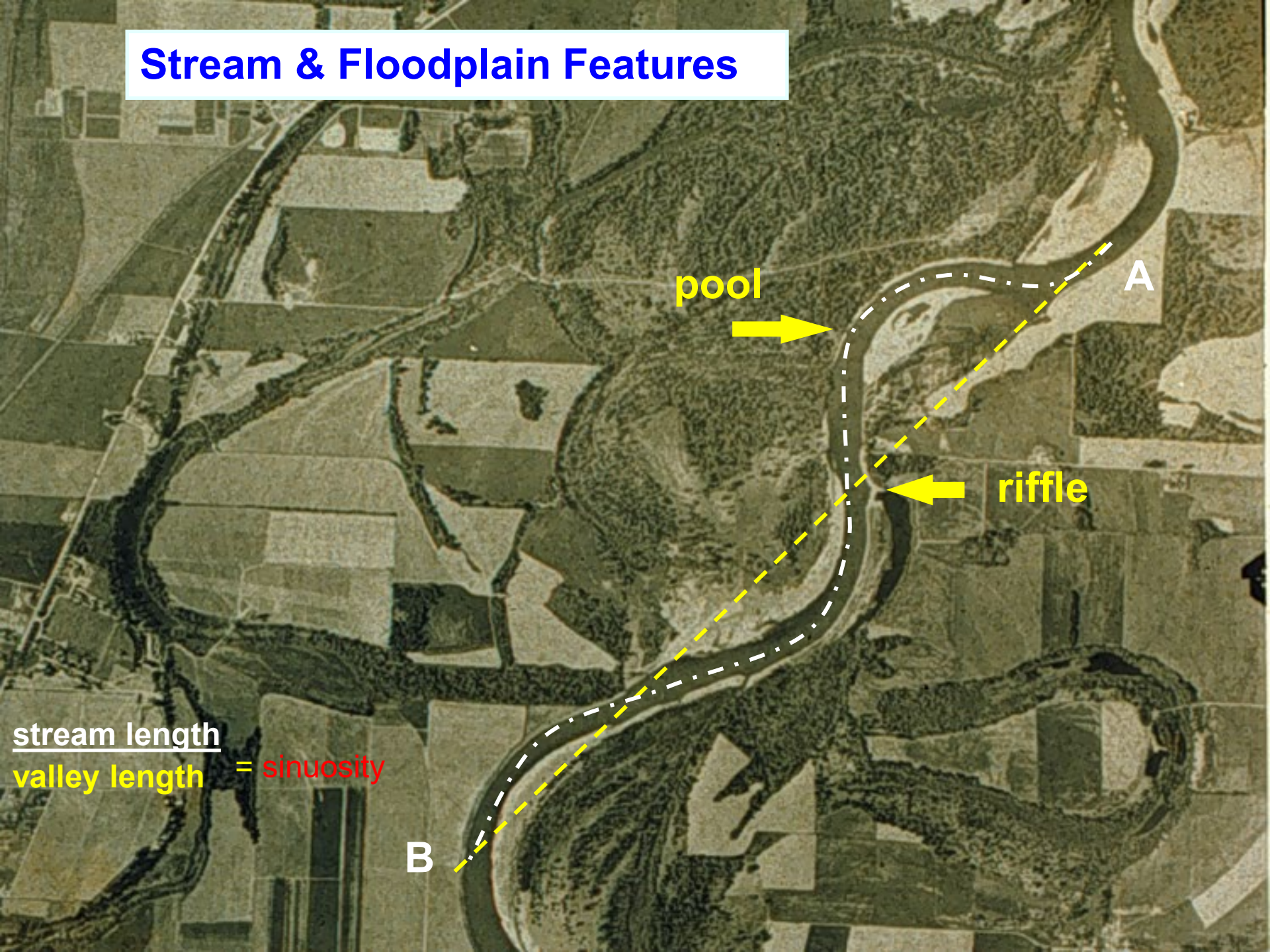
terrace

active channel

active floodplain



Stream & Floodplain Features



pool

A

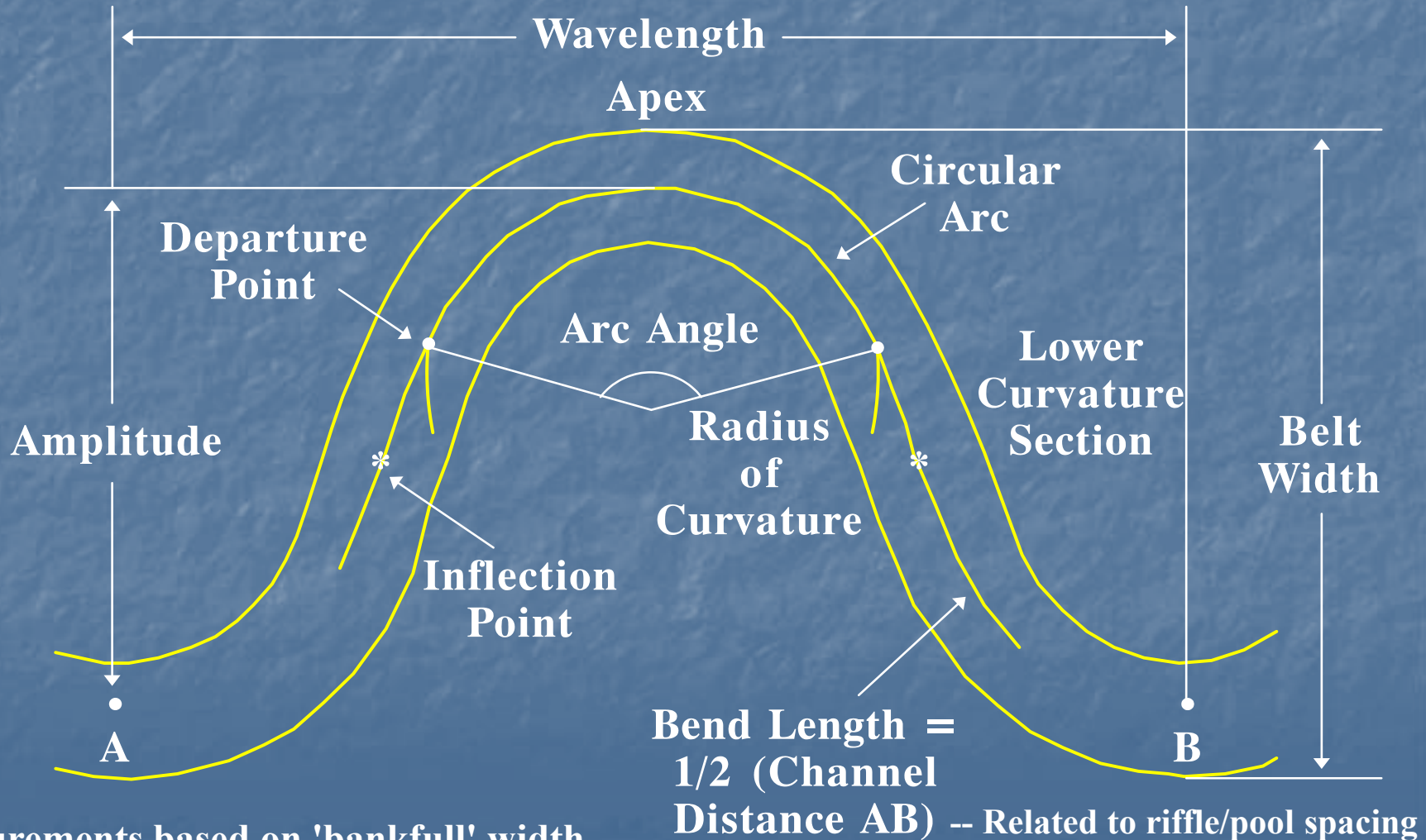
riffle

B

stream length
valley length = sinuosity

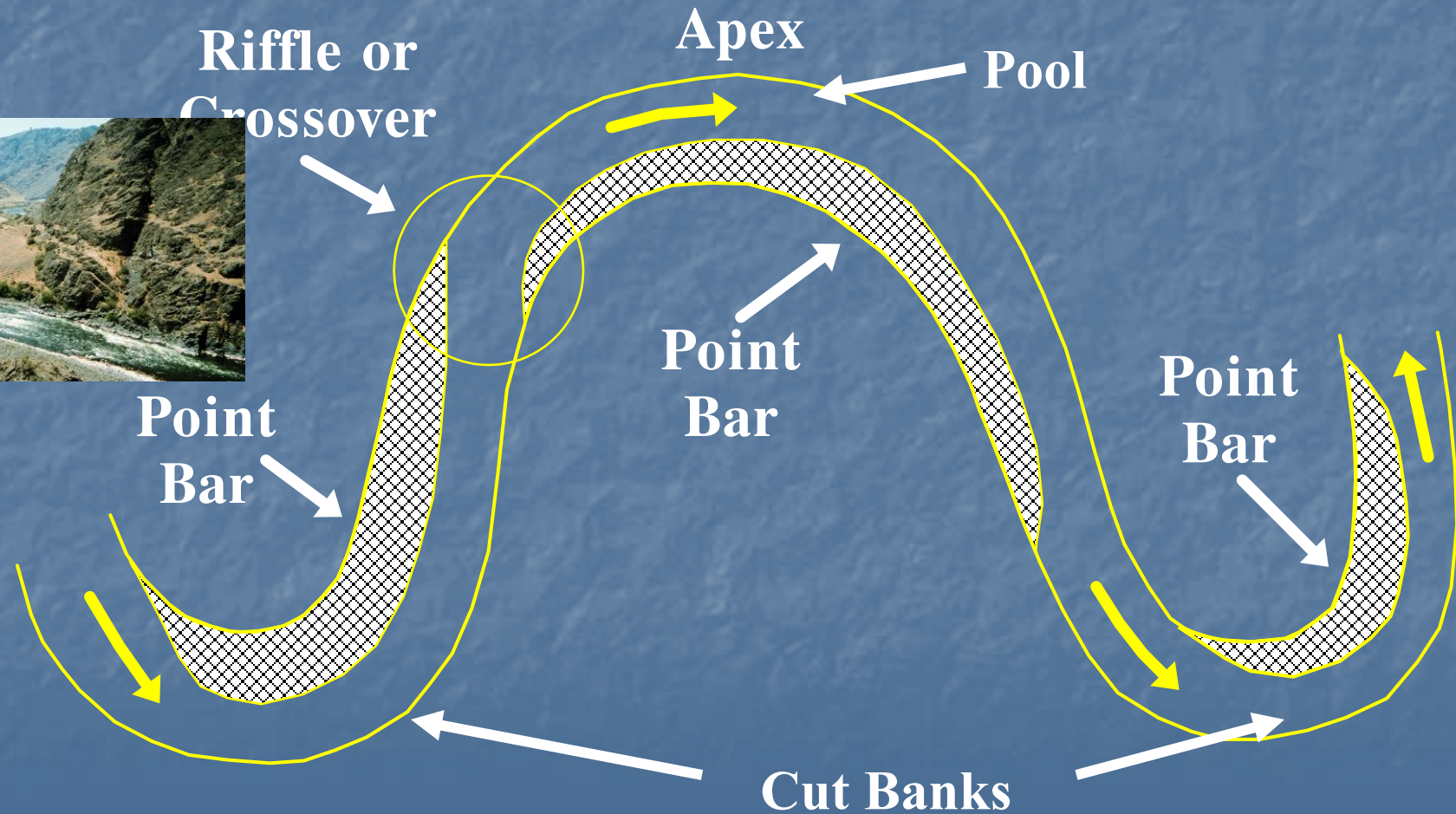
Plan View of a River Meander

(Leopold, et. al., 1964)



Point Bars In a River Meander

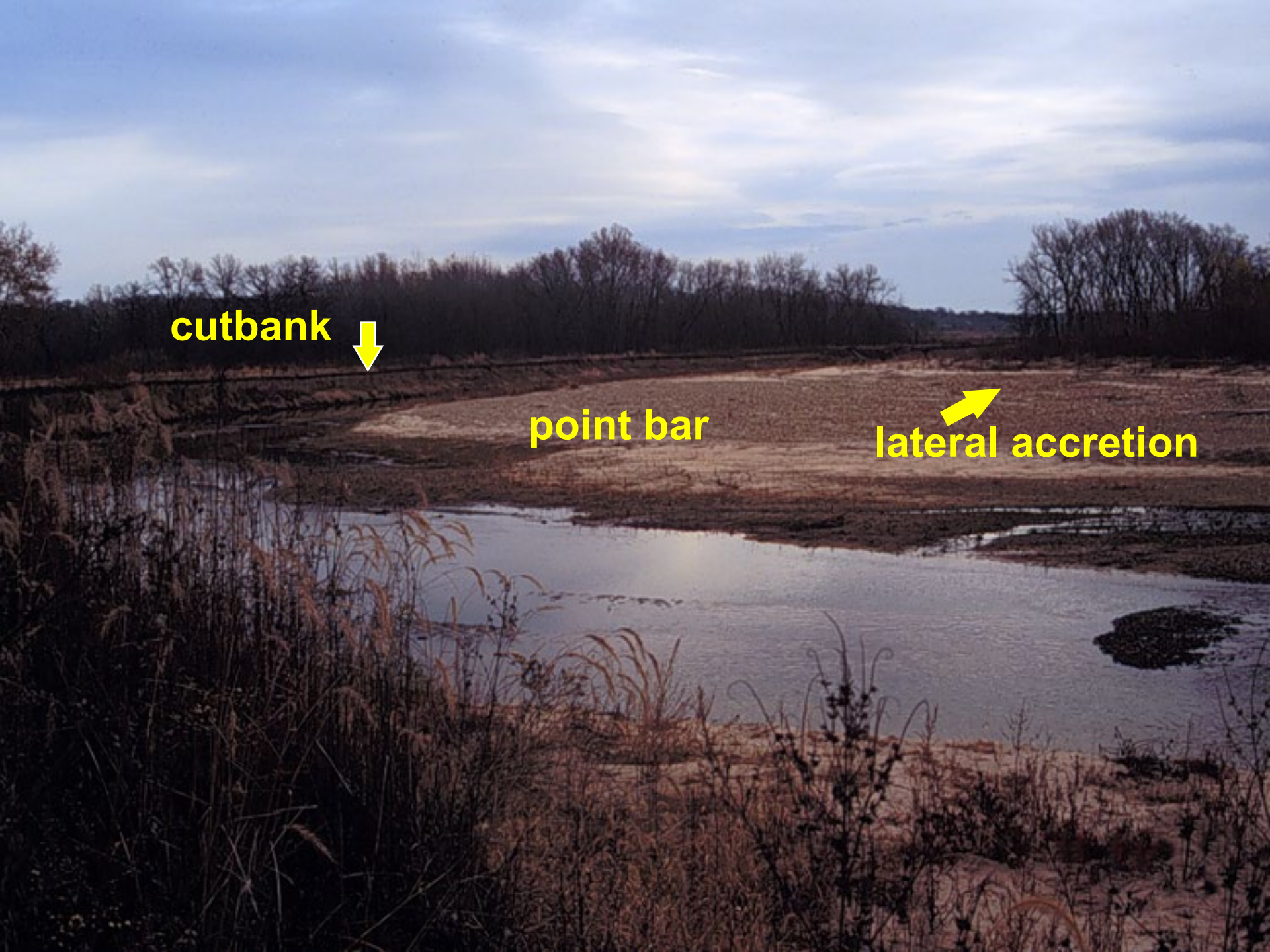
(Simons and Senturk, 1976)



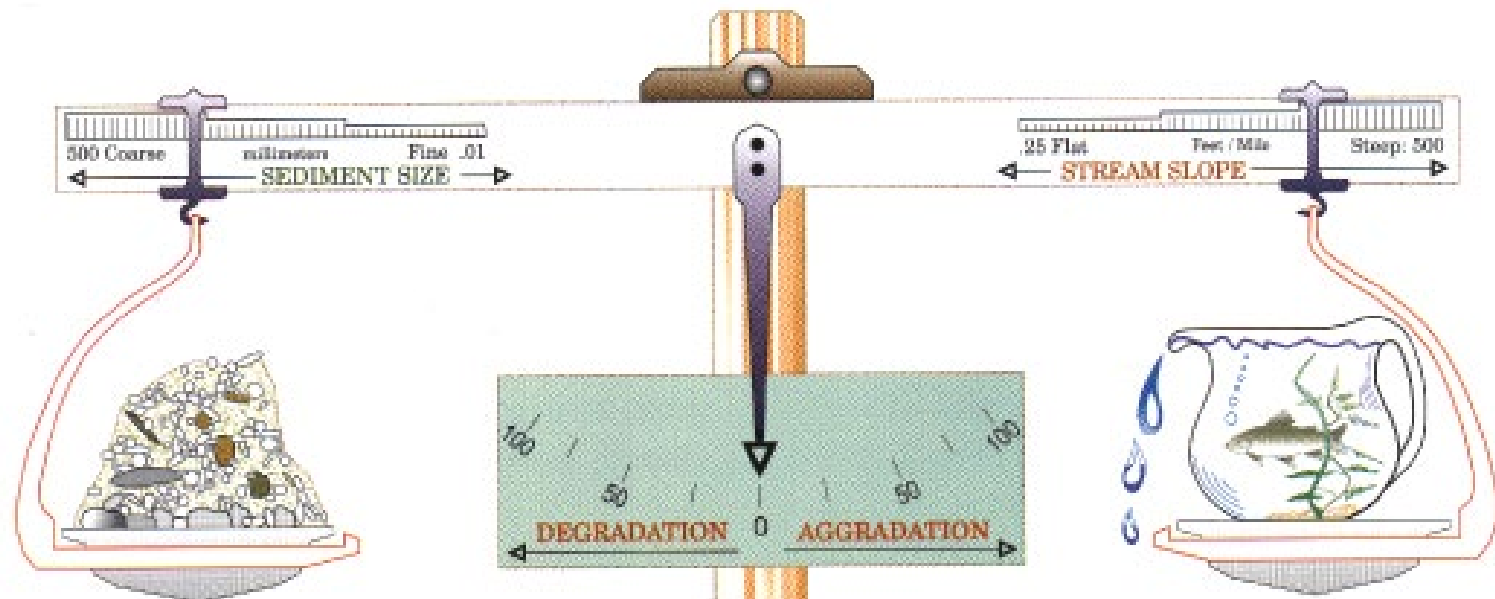
cutbank ↓

point bar

↗
lateral accretion



Lane's Model



Longitudinal profiles:

erosion

transition

deposition

$$Q_s \cdot D_{50} \propto S \cdot Q_w$$

(Sediment LOAD) x (Sediment SIZE) ∝ (Stream SLOPE) x (Stream DISCHARGE)



Walla Walla River, WA

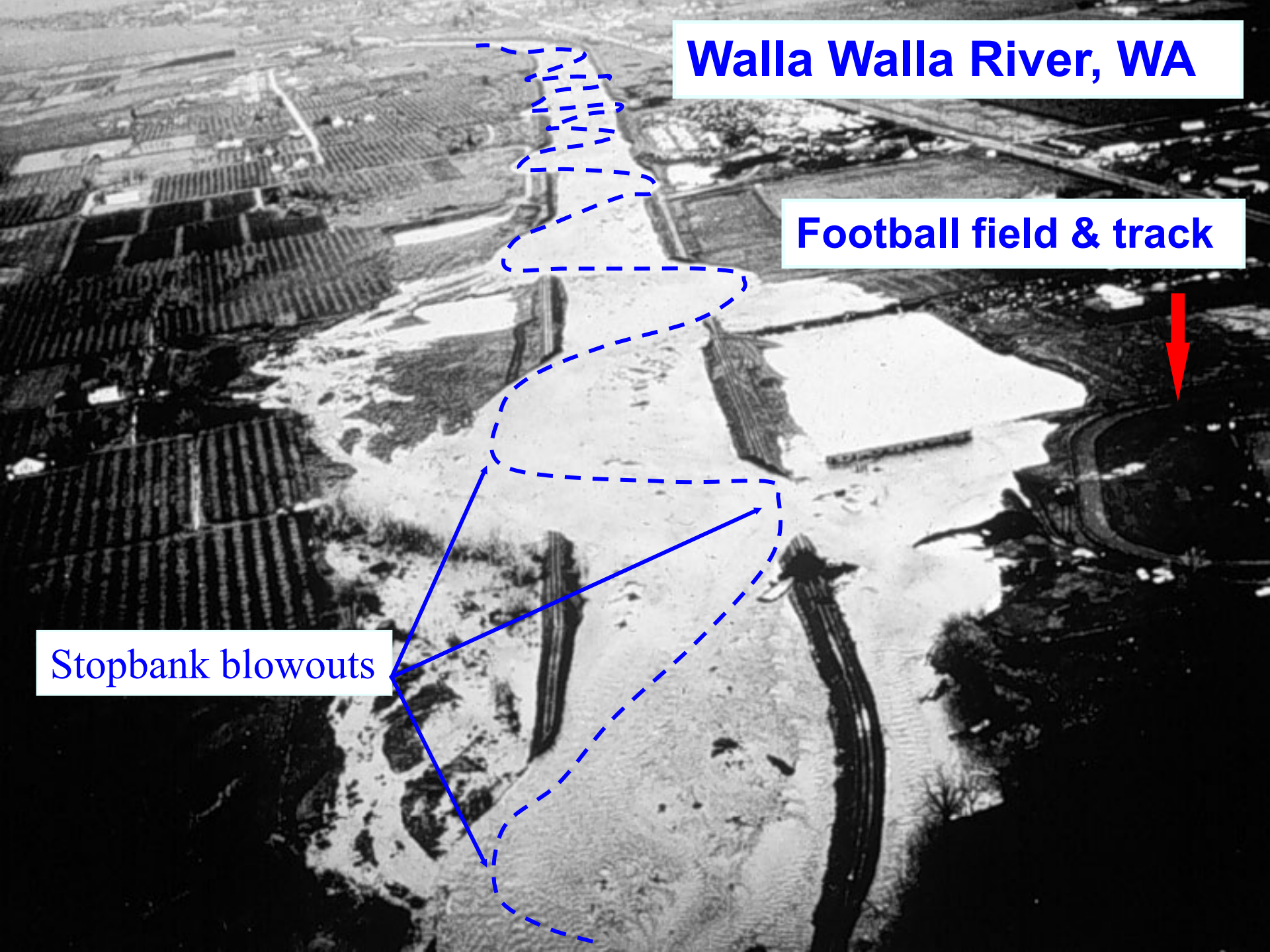
football field & track

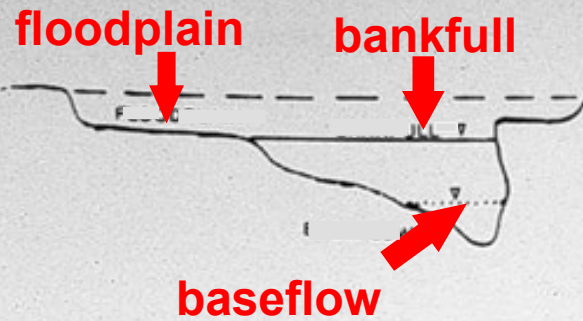
next slide view

Walla Walla River, WA

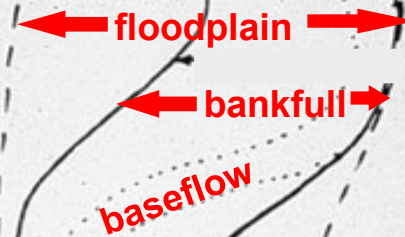
Football field & track

Stopbank blowouts

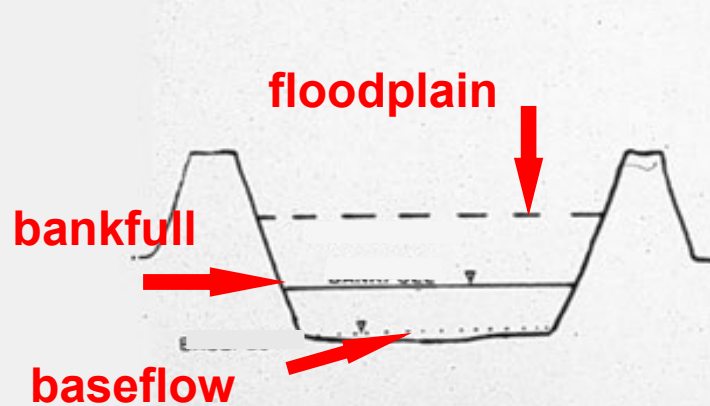




Natural Channel



Velocity?



Channelized "Design" Channel



Velocity?

Bank Failure

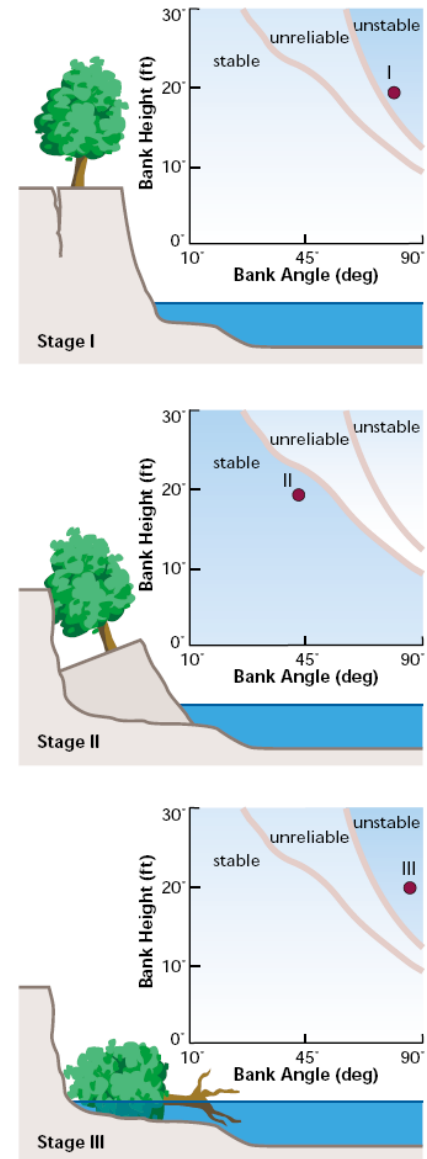


Figure 8.30: Bank failure stages. Stability of a bank will vary from stable to unstable depending on bank height, bank angle, and soil conditions.

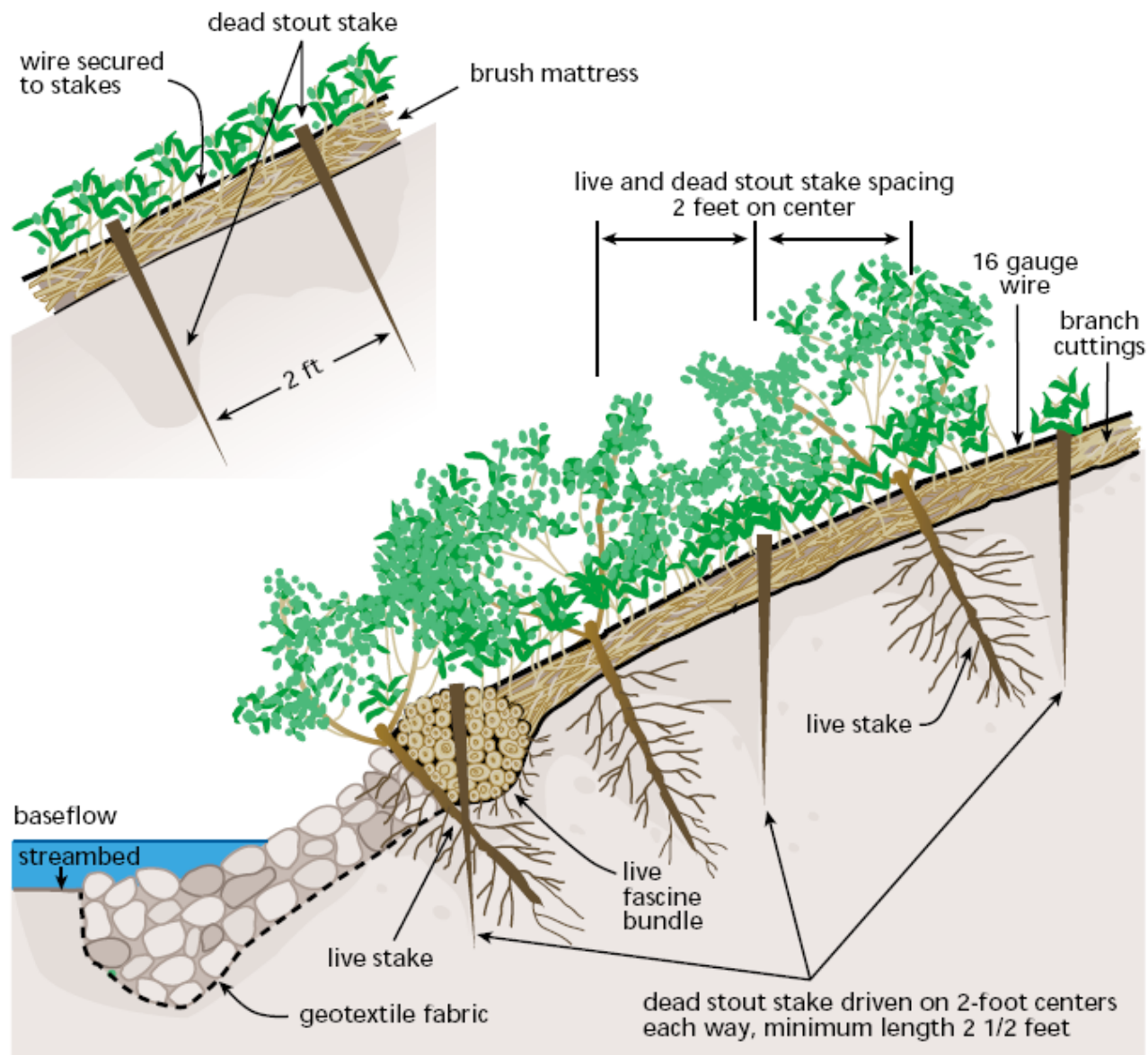
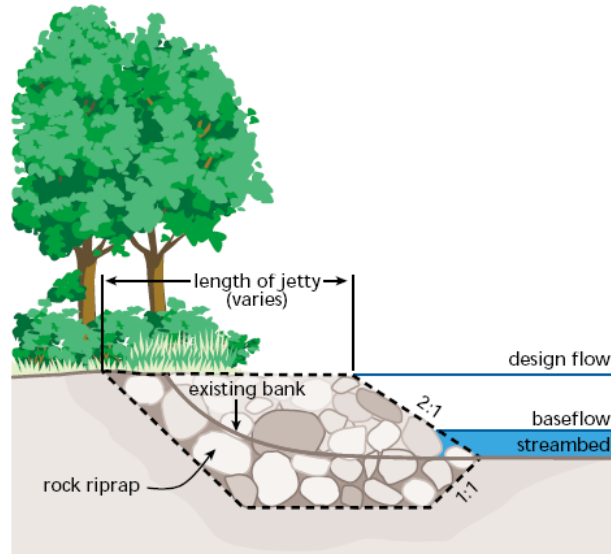


Figure 8.40: Cutting systems. Details of brushmattress technique.

Source: USDA-NRCS 1996a.

Note: Rooted/leafed condition of the living plant material is not representative at the time of installation.

Cross Section
not to scale



Front Elevation
not to scale

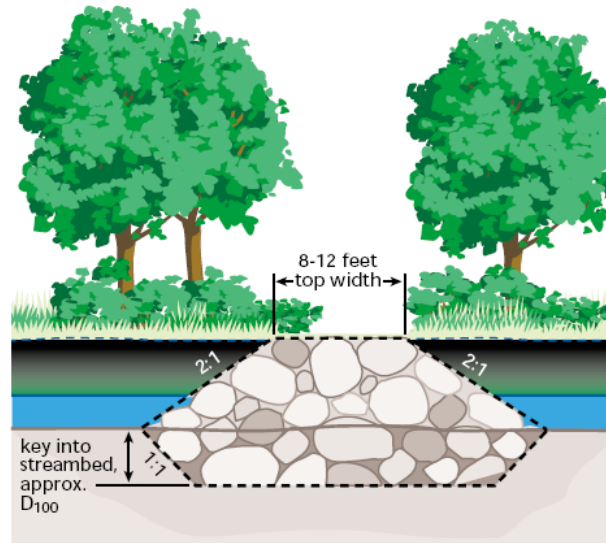
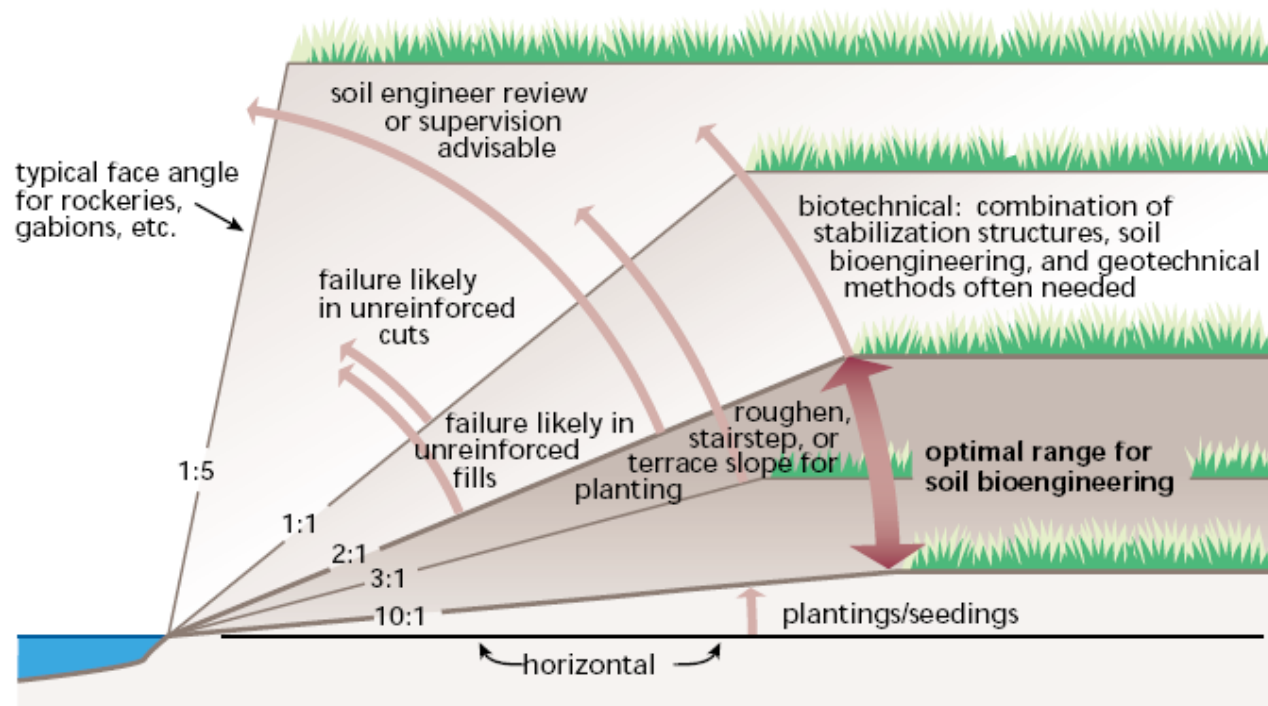


Figure 8.50: Instream habitat structure.

Wing deflector habitat structure.

Source: USDA-NRCS 1996a.



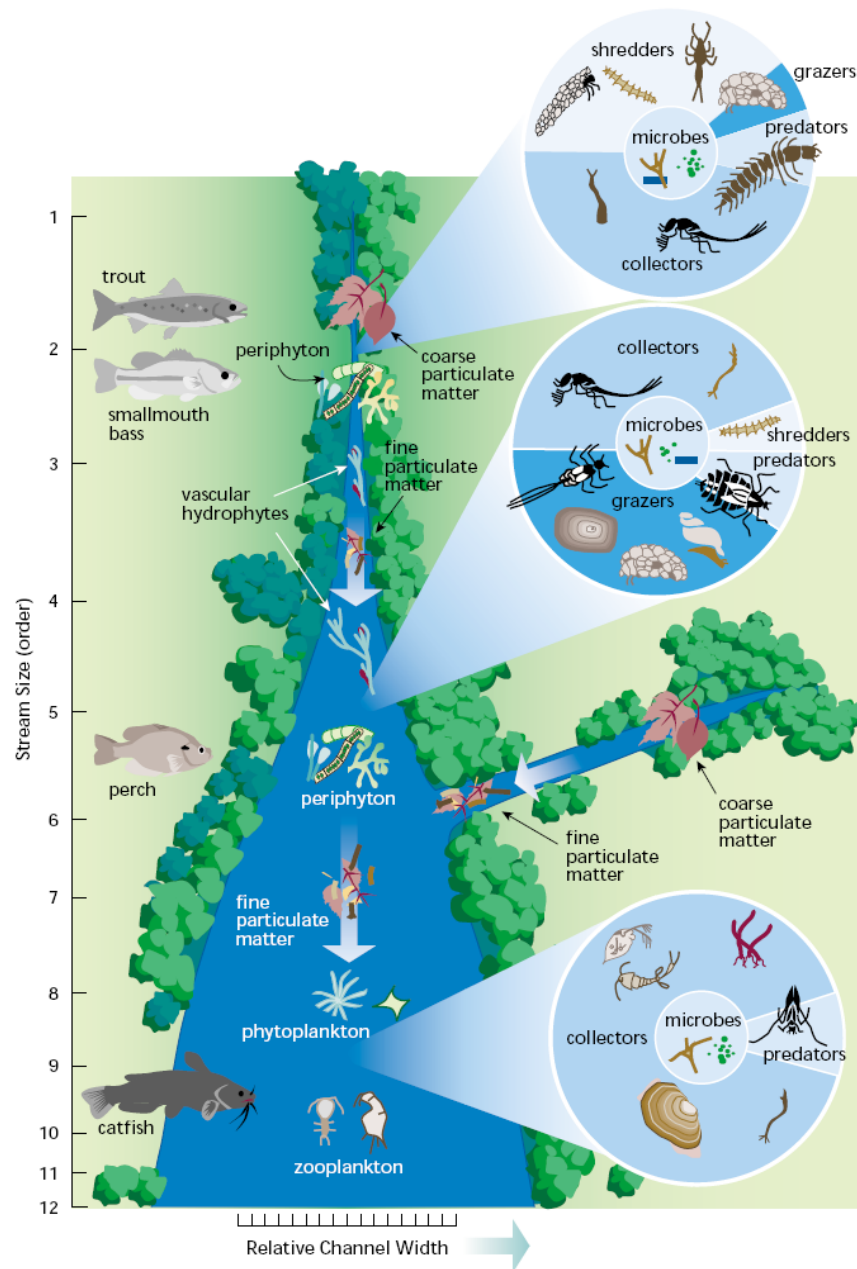


Figure 1.34: The River Continuum Concept. The concept proposes a relationship between stream size and the progressive shift in structural and functional attributes.

Source: Vannote et al. (1980). Published with the permission of NRC Research Press.

Restoration plans

- We need an integrated plan to restore rivers

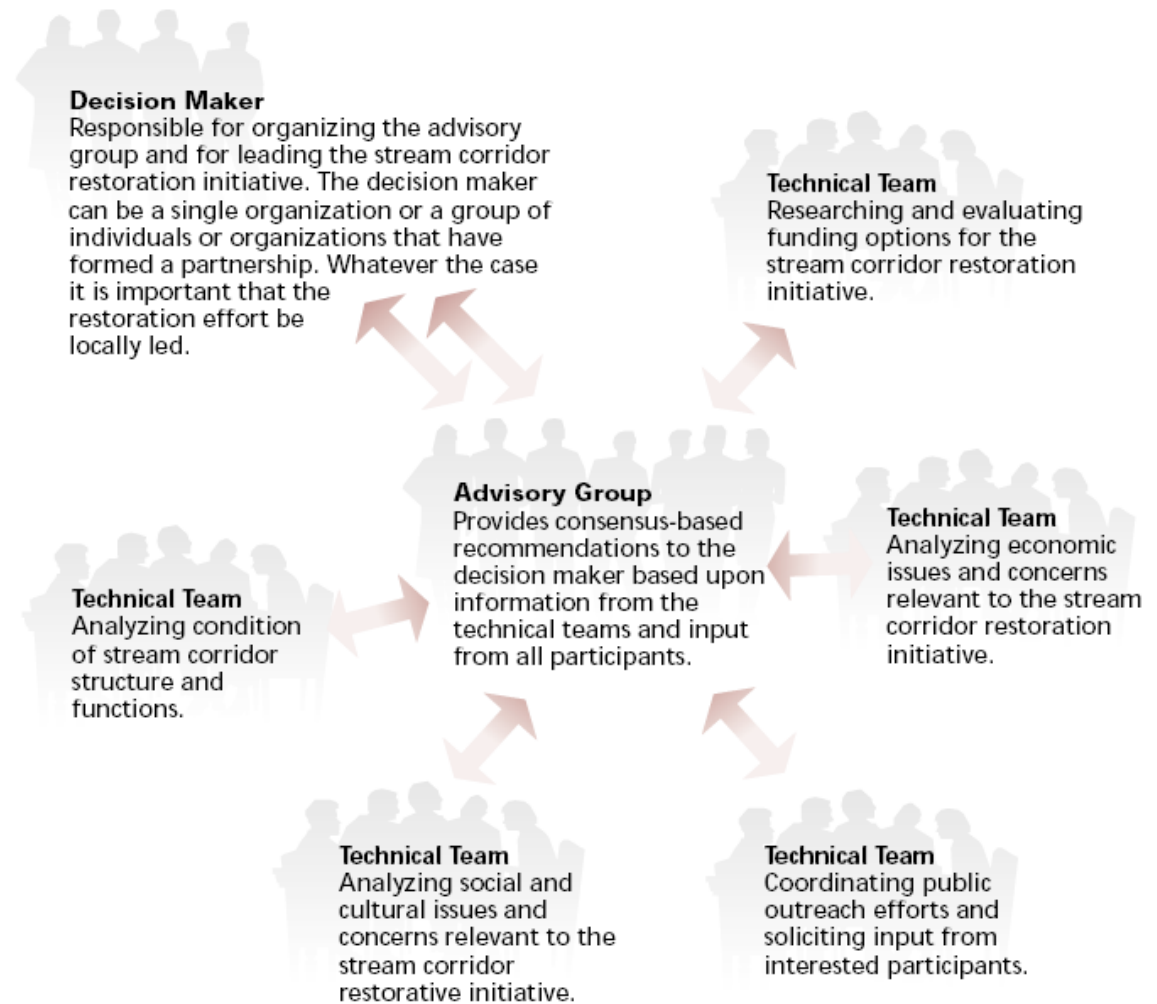


Figure 4.4: Flow of communication. Restoration plan development requires a decision structure that streamlines communication between the decision maker, the advisory group, and the various technical teams.

Options and costs

Solution	Level of Output		Cost (\$)		
	Total Output	Incremental Output	Total Cost	Incremental Cost	Incremental Cost Incremental Output
No action	0	0	0	0	0
A	80	80	2,000	2,000	25
B	100	20	2,600	600	30
E	120	20	3,600	1,000	50
F	140	20	7,000	3,400	170

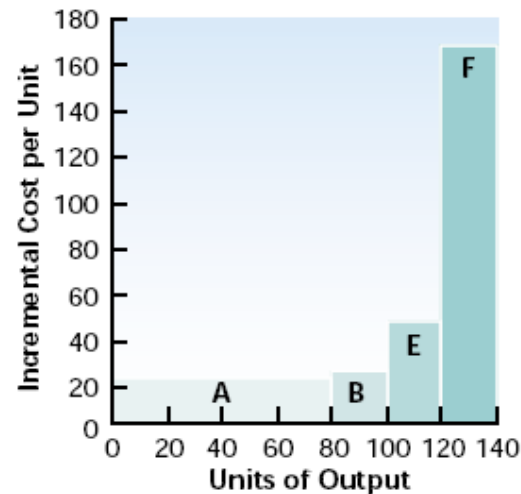


Figure 5.17: Incremental cost and output display. This graph plots the cost per unit (vertical axis) against the total output and incremental output (horizontal axis).

Reference Site and Monitoring

Primary Functions of Reference Sites

- *Can be used as models for developing restoration actions for a site.*
- *Provide a target to judge success or failure.*
- *Provide a control system by which environmental effects, unrelated to the restoration action, can be assessed.*

Basic Questions to Ask When Selecting Methods for Monitoring

- *Does the method efficiently provide accurate data?*
- *Does the method provide reasonable and replicable data?*
- *Is the method feasible within time and cost constraints?*

What to measure/monitor

- Evaluation Tools
- Criteria

General Objectives	Potential Evaluation Tools and Criteria
Channel capacity and stability	Channel cross sections
	Flood stage surveys
	Width-to-depth ratio
	Rates of bank or bed erosion
	Longitudinal profile
	Aerial photography interpretation
Improve aquatic habitat	Water depths
	Water velocities
	Percent overhang, cover, shading
	Pool/riffle composition
	Stream temperature
	Bed material composition
	Population assessments for fish, invertebrates, macrophytes
Improve riparian habitat	Percent vegetative cover
	Species density
	Size distribution
	Age class distribution
	Plantings survival
	Reproductive vigor
	Bird and wildlife use
Improve water quality	Aerial photography
	Temperature
	pH
	Dissolved oxygen
	Conductivity
	Nitrogen
	Phosphorus
	Herbicides/pesticides
	Turbidity/opacity
	Suspended/floating matter
Recreation and community involvement	Trash loading
	Odor
	Visual resource improvement based on landscape control point surveys
	Recreational use surveys
	Community participation in

Adaptive Management



- *Modify plans using monitoring, technical, and social feedback*
- *Track restoration policy, programs, and individual projects as feedback for further restoration policy and program redesign*
- *Restoration initiatives: recommend annual assessments*
 - *use monitoring data and other data/expertise*
 - *midcourse corrections or alternative actions*
 - *link reporting/monitoring schedules for midcourse corrections*
- *Manager may contract some/all monitoring, but periodically must visit sites, review reports, discuss with contractors.*

Mātauranga Māori in water infrastructure design

Refers to indigenous knowledge and wisdom from the Māori culture, and can significantly enhance water infrastructure design.

Here are some ways to incorporate it:

1.Holistic Approach: Integrating core water-sensitive design values with Mātauranga Māori and principles of tikanga Māori (traditional indigenous practices) provides a holistic, culturally enhanced approach to protecting water for future generations. [This approach aligns with natural hydrological water cycle processes and inherently provides for enhanced socio-cultural outcomes in addition to environmental stewardship¹.](#)

2.Recognition of Interconnectedness: Māori culture recognizes that environmental care is intrinsically linked with the mauri (life force) of the environment. Te Ao Māori (Māori world view) emphasizes the relationship between people, water, and the natural environment. By incorporating these perspectives, water infrastructure designs can better reflect the interconnectedness of ecosystems and human well-being.

3.Whakataukī (Proverbs): Whakataukī, or proverbs, play a significant role in Māori culture. For example:

- *“Ka ora te wai, Ka ora te whenua. Ka ora te whenua, Ka ora te tangata.”* (If the water is healthy, the land will be nourished. If the land is nourished, the people will be provided for.) This proverb highlights the interdependence of water, land, and people.

Mātauranga Māori in water infrastructure design

4. Community Engagement: Engage with local Māori communities during the design process. Seek their input, listen to their perspectives, and incorporate their knowledge into the infrastructure design. This collaborative approach ensures that the design respects cultural values and aspirations.

5. Incorporate Traditional Practices: Consider incorporating traditional practices related to water management, such as:

- **Riparian Planting:** Planting native vegetation along waterways to improve water quality and habitat.
- **Large Wood Placement:** Strategically placing large wood (fallen trees, branches) in streams to enhance habitat complexity, stabilize banks, and improve water quality.

6. Research and Case Studies: Explore research and case studies that demonstrate successful integration of Mātauranga Māori in water-sensitive urban design. [These examples can provide practical insights and inspire innovative approaches²³](#).

- Remember that each context is unique, and collaboration with local communities is essential for meaningful incorporation of Mātauranga Māori into water infrastructure design. By doing so, we can create more sustainable, culturally respectful, and resilient solutions for our water systems.